

Offline training path

Trace capture

Record page statistics, attention traces, and decode outcomes.

Oracle labels

Choose the safest low-cost page format from replay data.

Selector dataset

Assemble features, labels, and operating-point targets.

Train selector

Fit a lightweight classifier for quality and systems profiles.

Trained selector

Exported policy checkpoint

Runtime serving path

Runtime selector

Score each new KV page using the trained serving policy.

Page-format write path

Assign M0 / M3-style formats before pages land in cache.

Compressed KV pages

Store page-organized representations as the execution format.

Fused score + mix

Attend directly on compressed pages with no dense widening.

serving-time attention
Decode output
Next token and
cache update